

Isentropic Temp Relation

We are
deriving
this

$$\frac{T_0}{T} = 1 + \frac{\gamma-1}{2} M^2$$

② We want to connect to M (Mach)

i) Divide by T

$$\frac{T_0}{T} = \frac{T}{T} + \frac{V^2}{2c_p T}$$

ii) T_0/T is a ratio

"How much is total temp compared to static temp?"

iii) Result

$$\underline{\underline{\frac{T_0}{T} = 1 + \frac{V^2}{2c_p T}}}$$

① Big Picture

Total Temp = static temp
+ motion-energy contribution

$$\text{Formula: } \underline{\underline{T_0 = T + \frac{V^2}{2c_p}}}$$

③ Bring in Mach!

$$i) M = \frac{V}{a} \rightarrow V = Ma$$

$$\underline{\underline{V^2 = M^2 a^2}}$$

④ Bring in speed of sound!

$$i) a = \sqrt{\gamma RT} \rightarrow a^2 = \gamma RT$$

ii) Sub into $V^2 = M^2 a^2$

$$\underline{\underline{V^2 = M^2 (\gamma RT)}}$$

This connects velocity
energy to
temperature

⑤ Substitute V^2

$$\frac{T_0}{T} = 1 + \frac{V^2}{2c_p T} \rightarrow \frac{T_0}{T} = 1 + \frac{M^2 \gamma R \cancel{T}}{2c_p \cancel{T}} \rightarrow \underline{\underline{\frac{T_0}{T} = 1 + \frac{M^2 \gamma R}{2c_p}}}$$

Static Temp cancels b/c we're already using a temp. ratio

⑥ Use ideal-gas heat capacity generation

i) For an ideal gas: $c_p = \frac{\gamma R}{\gamma - 1} \rightarrow \frac{\gamma R}{c_p} = \gamma - 1$

ii) Substitute: $\frac{T_0}{T} = 1 + \frac{M^2 \gamma R}{2c_p}$, $\frac{\gamma R}{c_p} = \gamma - 1$

$$\boxed{\frac{T_0}{T} = 1 + \frac{M^2 (\gamma - 1)}{2}}$$

What does this mean?

The ratio between stagnation temperature and static temperature depends on the Mach number.

Interviewer : Okay, can you derive the relationship between stag. temp & static temp. as a function of Mach number ?

1) Start with $\rightarrow T_0 = T + \frac{V^2}{2C_p}$ 2) $\frac{T_0}{T} = 1 + \frac{V^2}{2C_p T}$

3) $M = \frac{V}{a} \rightarrow V = Ma \rightarrow V^2 = M^2 a^2$ 4) $a = \sqrt{\gamma R T} \rightarrow a^2 = \gamma R T$

5) $V^2 = M^2 \gamma R T$ 6) $\frac{T_0}{T} = 1 + \frac{M^2 \gamma R T}{2C_p T} \rightarrow \frac{T_0}{T} = 1 + \frac{M^2 \gamma R}{2C_p}$

7) ideal gas heat capacity generation : $C_p = \frac{\gamma R}{\gamma - 1} \rightarrow \frac{\gamma R}{C_p} = \gamma - 1$

8) $\boxed{\frac{T_0}{T} = 1 + \frac{M(\gamma - 1)}{2}}$

As the ratio increases,
so does the mach #
and vice versa !

$$M = \frac{V}{a} \rightarrow V = Ma \rightarrow V^2 = M^2 a^2$$

$$a = \sqrt{\gamma R T} \rightarrow a^2 = \gamma R T$$

$$V^2 = M^2 [\gamma R T]$$

$$T_0 = T + \frac{V^2}{2C_p} \rightarrow \frac{T_0}{T} = 1 + \frac{V^2}{2C_p T}$$

$$\frac{T_0}{T} = 1 + \frac{M^2 \cancel{\gamma R T}}{2C_p \cancel{T}} \rightarrow \frac{T_0}{T} = 1 + \frac{M^2 \gamma R}{2C_p}$$

$$C_p = \frac{\gamma R}{\gamma - 1} \rightarrow \frac{\gamma R}{C_p} = \gamma - 1$$

$$\boxed{\frac{T_0}{T} = 1 + \frac{M^2 (\gamma - 1)}{2}}$$